Exothermic and Endothermic Reactions

Exothermic	Endothermic

Materials for Each Group

- Vinegar
- Baking soda
- Calcium chloride
- Water
- Thermometer
- 4 small clear plastic cups
- 1 cup measuring cup
- Measuring spoons (1 tablespoon, ½ teaspoon)

Baking Soda and Vinegar

1. Pour about 10 mL of vinegar into a small plastic cup. Then, place a thermometer into the vinegar. Record the initial temperature (T_i) in the table below.
2. While the thermometer is in the cup, add about ½ teaspoon of baking soda to the cup.
3. Watch the thermometer for any change in temperature. After it has stopped changing, record the final temperature (T_f) and any other observations you made in the table below.

Baking Soda and Calcium Chloride

1. Make a baking soda solution by dissolving about 2 tablespoons of baking soda in 1 cup of water. Stir until no more baking soda will dissolve.
2. Place about 10 mL of baking soda solution in a small plastic cup. Then, place a thermometer into the baking soda solution. Record the initial temperature (T_i) in the table below.
3. While the thermometer is in the cup, add ½ teaspoon of calcium chloride to the cup.
4. Watch the thermometer for any change in temperature. After it has stopped changing, record the final temperature (T_f) and any other observations you made in the table below.

Process	**T_i**	**T_f**	**Change in Temperature** **(Final temp – initial temp**	**Exothermic or Endothermic ?**	**Other observations?**
Baking soda + vinegar					
Baking soda solution + calcium chloride					